

Roads, Oil and Local Economic Activity in Azerbaijan

Spatial Evidence from Nighttime Lights

Elvin Mammadov

TU Dresden

Pitch Presentation

Research question

Do oil-price shocks affect local economic activity differently across space in Azerbaijan?

- Azerbaijan is highly dependent on oil revenues.
- Investment and infrastructure may be spatially concentrated around Baku.
- Nighttime lights allow local economic activity to be measured at high spatial resolution.

Literature Foundation

Storeygard (2016) shows that transport infrastructure affects local economic activity using spatial variation and satellite data. Following this logic, this study combines oil shocks with geographic exposure to analyze heterogeneous regional effects in Azerbaijan.

Outcome Variable

- SVNL dataset (Chen et al. 2024, Scientific Data) 1992–2023, 500m resolution, via Google Earth Engine
- Unit of analysis: pixel $\times$ year (2010–2020)

Road Exposure

- Major and secondary roads (OpenStreetMap)
- Distance computed from pixel centroid

Investment Measure

- Total fixed capital investment
- Source: State Statistical Committee
- Unit of analysis: district $\times$ year

Identification

- National oil shocks & distance to Baku
- Main interaction: $\text{Oil}_t \times \text{Distance}_i$

Data Sources

Source	Variable	Unit
Google Earth Engine SVNL (Chen et al. 2024)	Nighttime lights (NTL)	Pixel-year
OpenStreetMap	Road network	Pixel
Statistical Committee of Azerbaijan	Fixed capital investment	District-year
World Bank / Brent	Oil price	Year
GADM	Administrative boundaries	District

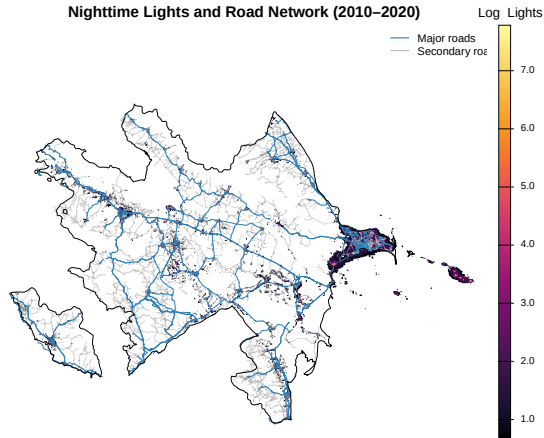
Descriptive Statistics: Pixel Level

Final sample

2010–2020, fixed pixel-year panel

Variable	N	Mean	SD	Min	Median	Max
NTL	412,797	3.9	12.0	0.0	1.6	1,365.8
Log(NTL+1)	412,797	1.1	0.9	0.0	0.9	7.2
Brent oil price	412,797	76.2	26.1	42.3	71.1	112.0
Distance to Baku (km)	412,797	150.6	103.4	0.2	150.6	416.6
Distance to major road (m)	412,797	2,859.0	3,474.9	0.0	1,565.6	32,282.9
Distance to secondary road (m)	412,797	1,033.6	1,373.9	0.0	564.1	14,687.8

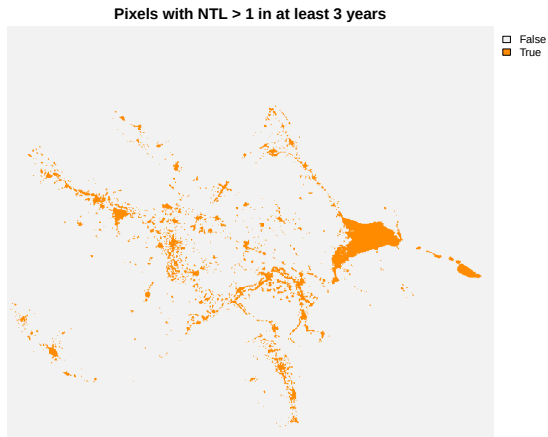
Spatial Data: Network Overview



Maximum Nighttime lights values of each pixel
and road network in Azerbaijan

Major roads: Motorway, trunk, and primary roads. **Secondary roads:** Secondary and tertiary roads.

Spatial Data: Persistent Pixels

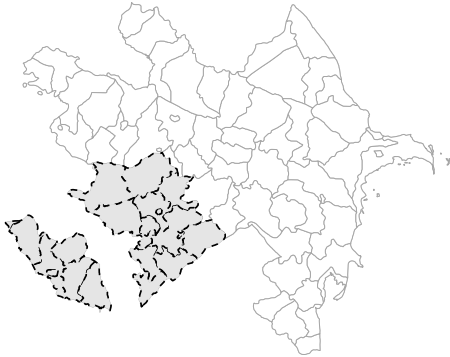


Only the pixels which the value is more than 1 at least in 3 years are included

Spatial Data: Persistent Pixels

Study Area and Excluded Regions

Nakhchivan and Karabakh-related districts are excluded from the main sample



Main sample excludes Nakhchivan and Karabakh-related districts due to geographic separation and the structural break after the 2020 war.

Reduced-form specification

$$\log(NTL_{it} + 1) = \beta (\text{Oil}_t \times \log(\text{DistanceBaku}_i)) + \alpha_i + \lambda_t + \varepsilon_{it}$$

- $\log(NTL_{it} + 1)$: log nighttime lights in pixel i and year t
- Oil_t : annual Brent oil price
- DistanceBaku_i : distance from pixel i to Baku
- α_i : pixel fixed effects controlling for time-invariant local characteristics
- λ_t : year fixed effects controlling for nationwide annual shocks
- Expected sign: $\beta < 0$, meaning that pixels farther from Baku benefit less from oil-price increases

Descriptive Statistics: District Level

Investment and District Sample

District-year panel observations ($N = 605$)

	N	Mean	SD	Min	Median	Max
Total fixed capital investment	605	265034.8	1404191.1	2314.2	52235.3	11868970.6
Log investment	605	10.9	1.2	7.7	10.9	16.3
Brent oil price	605	76.2	26.1	42.3	71.1	112.0
Distance to Baku (km)	605	200.8	88.0	30.4	187.0	396.9
Oil $\times$ log distance to Baku	605	395.6	142.0	145.8	369.9	670.2

Mechanism Check: District Investment

Investment response to oil-price shocks

$$\log(\text{Investment}_{rt}) = \gamma (\text{Oil}_t \times \log \text{DistanceBaku}_r) + \mu_r + \lambda_t + u_{rt}$$

- Investment is measured at the district-year level.
- District and year fixed effects control for time-invariant district differences and national annual shocks.
- The estimated coefficient is positive but statistically weak.
- Therefore, investment is interpreted as suggestive mechanism evidence, while the main analysis relies on reduced-form effects on nighttime lights.

Reduced-Form Results: Oil Shocks and Nighttime Lights

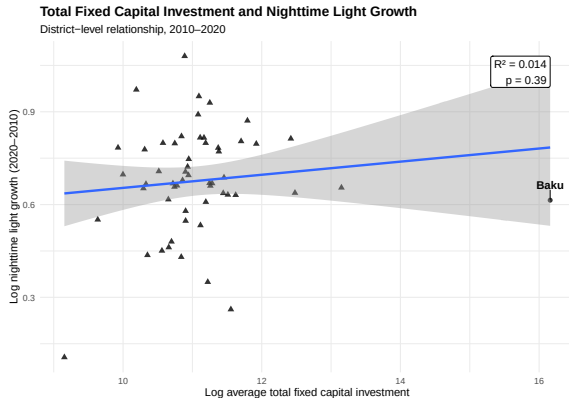
	Baseline Reduced Form	Road-Adjusted Specification	Investment Mechanism
Brent oil price $\times$ log dist. to Baku	-0.0003 (0.0002)	-0.0033*** (0.0006)	0.0043 (0.0031)
Log dist. to major road		-0.1311*** (0.0187)	
Log dist. to secondary road		-0.1557*** (0.0336)	
Observations	412797	412797	605
R ²	0.884	0.501	0.768
Within R ²	0.0007	0.2099	0.0114
Pixel fixed effects	X		
Year fixed effects	X	X	X
District fixed effects		X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Dep. vars: Cols 1–2 $\log(NTL + 1)$, Col 3 $\log(\text{Investment}_{it})$

Interpretation: Negative coefficients suggest that pixels farther from Baku benefit less from oil-price increases, but the baseline estimate is statistically weak.

Investment and Nighttime Light Growth



District-level relationship between total fixed capital investment and nighttime-light growth.

Limitations

- Investment data are available only at district level.
- Fixed capital investment includes both public and private sources.
- First-stage relationship is weak.
- Road construction dates are unavailable.
- Karabakh-related structural break requires sample restrictions.

Thank you!